GRPO unraveled: A Small-Scale Empirical Study of RLVR
CMPT 981 Project Milestone

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Abstract

[TODO: Abstract. Write last: the project goal in one sentence, the three papers reviewed, and the progress made since the proposal.]

1 Introduction

[TODO: Introduction]

2 Paper Reviews

[TODO: Shared notation (a few lines)]

2.1 Success amplification [Mroueh, 2025]

Setting and assumptions. [TODO: Setting and assumptions]

Main results. [TODO: Main results]

Connection to our project. [TODO: Connection to our project]

Limitations and open questions. [TODO: Limitations and open questions]

2.2 Gradient gap and step-size thresholds [Suk and Duan, 2025]

Setting and assumptions. [TODO: Setting and assumptions]

Main results. [TODO: Main results]

Connection to our project. [TODO: Connection to our project]

Limitations and open questions. [TODO: Limitations and open questions]

2.3 Pass@k advantage shaping [Thrampoulidis et al., 2026]

Setting and assumptions. Thrampoulidis et al. [2026] study how advantage shaping changes the objective optimized by RLVR methods. They consider binary rewards, where a correct response receives 1 and 0 otherwise. Under their setting, for each prompt x, the model independently samples

a training group of G responses from its current policy. If its success probability is $p_\theta(x)$, the probability of obtaining at least one correct response in k independent attempts is

$$P_k(x) = 1 - (1 - p_\theta(x))^k.$$

Here, G is the number of responses during the training phase, while k is the number of attempts considered by the pass@k evaluation metric. Their analysis assumes one policy update per freshly sampled training group, with clipping inactive and KL regularization omitted.

Main results. The gradient of the $P_k(x)$, pass@k objective, is

$$\nabla_\theta P_k(x) = k(1 - p_\theta(x))^{k-1} \nabla_\theta p_\theta(x).$$

For fixed $k > 1$, a high success probability $p_\theta(x)$ makes $1 - p_\theta(x)$ really small. So pass@k optimization places less weight on prompts already solved reliably, and more weight to prompts with lower success probabilities.

To estimate the pass@k gradient from sampled responses, the authors construct RLOO_k, an unbiased estimator that uses the other responses in each group to calculate leave-one-out weights. Applying standard deviation normalization produces their biased GRPO_k variant. More broadly, the paper connects advantage shaping to surrogate reward maximization, explaining how existing shaping rules can be interpreted through the objective they optimize.

Connection to our project. For our experiment, we are aiming to compare GRPO with biased and unbiased pass@k estimators on a simple synthetic task that has one correct answer. To do this, we will vary the training group size G, while holding the total sampling budget fixed. We will then measure how each trained model performs on different pass@k versions. Our goal here is to determine whether larger training groups improve performance under the same sampling budget. Moreover, under these settings, we can also compute population gradients directly and use it to compare them with average sampled gradients during training.

Limitations and open questions. Thrampoulidis et al. [2026] note that their surrogate reward interpretation relies on an infinite group limit with k being fixed. But for practical perspectives and also in our experiments, we use finite groups instead, where noisy success rate estimates can cause updates to differ from the theoretical prediction. This raises a question about how closely training with small groups follows the population predictions.

References

Youssef Mroueh. Reinforcement learning with verifiable rewards: GRPO's effective loss, dynamics, and success amplification. *arXiv preprint arXiv:2503.06639*, 2025.

Joe Suk and Yaqi Duan. On the optimization dynamics of RLVR: Gradient gap and step size thresholds. *arXiv preprint arXiv:2510.08539*, 2025.

Christos Thrampoulidis, Sadegh Mahdavi, and Wenlong Deng. Advantage shaping as surrogate reward maximization: Unifying pass@K policy gradients. *Transactions on Machine Learning Research*, 2026. arXiv:2510.23049.